



FINAL YEAR ENGINEERING PROJECT

Smart Medicine Availability & Intelligent Janaushadhi Recommendation System

DESIGN SYSTEM

Project	Smart Medicine Availability & Intelligent Janaushadhi Recommendation System
Document	DESIGN SYSTEM
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Smart Medicine System — Visual Design Language

1. What is a Design System?

A design system is a set of rules, tokens, and components that define how the application looks and feels. It ensures:

- Every page has a consistent visual style
- Developers don't invent their own colours, fonts, or spacing
- The UI feels professional and trustworthy (critical for healthcare)

This project's design system is implemented in `src/index.css` using Tailwind v4's `@theme {}` CSS-first approach.

2. Healthcare Theme

The visual identity was designed specifically for a **healthcare application**:

- **Medical Blue** (`#2563eb`) — primary actions, links, trust
- **Teal** (`#0d9488`) — secondary actions, healthcare accent
- **Clean White** backgrounds — medical/clinical feel
- **Slate greys** — professional, neutral text
- **Green** for success (In Stock), **Red** for danger (Out of Stock)

This is intentional: healthcare apps should feel **safe, professional, and trustworthy**. Playful or loud colours would undermine user confidence.

3. Colour Palette

Primary — Medical Blue

Token	Hex	Usage
primary-50	<code>#eff6ff</code>	Light background tints
primary-100	<code>#dbeafe</code>	Badge backgrounds
primary-300	<code>#93c5fd</code>	Placeholder icons
primary-600	<code>#2563eb</code>	**Main buttons, links**
primary-700	<code>#1d4ed8</code>	Button hover
primary-900	<code>#1e3a8a</code>	Dark headings

Secondary — Teal

Token	Hex	Usage
secondary-500	<code>#14b8a6</code>	Secondary buttons, accents

Token	Hex	Usage
secondary-600	#0d9488	Secondary hover

Semantic Colours

Colour	Variant	Usage
Green	success-*	In Stock, positive feedback
Amber	warning-*	Limited Stock, low inventory
Red	danger-*	Out of Stock, delete, errors
Blue	info-*	Jan Aushadhi badges, information
Purple	accent-*	New arrivals, highlights

Text Colours

text-slate-900 → Primary headings
 text-slate-700 → Body text
 text-slate-500 → Secondary text
 text-slate-400 → Muted/helper text
 text-slate-300 → Placeholders

4. Typography

Font: Inter — chosen for its exceptional legibility in data-dense UI.

Type Scale

Class	Size	Usage
text-[10px]	10px	Ribbon labels, tiny tags
text-xs	12px	Helper text, captions
text-sm	14px	Body text, card content
text-base	16px	Default body
text-lg	18px	Subtitles
text-xl	20px	Section headings
text-2xl	24px	Page sub-headings
text-3xl	30px	Page titles
text-4xl	36px	Hero headings
text-5xl	48px	Display / hero

Font Weights

Class	Weight	Usage
font-normal	400	Body text
font-medium	500	Labels, secondary headings

Class	Weight	Usage
font-semibold	600	Buttons, strong labels
font-bold	700	Card titles, page headings
font-extrabold	800	Hero text, prices

5. Spacing System

Tailwind's 4px base unit:

- **p-1** = 4px, **p-2** = 8px, **p-3** = 12px, **p-4** = 16px, **p-6** = 24px, **p-8** = 32px

Standard patterns used in this project:

Card padding:	p-4 (16px) or p-6 (24px)
Section padding:	py-20 (80px vertical)
Component gap:	gap-3 (12px) or gap-4 (16px)
Badge gap:	gap-1.5 (6px)
Form field margin:	mb-4 (16px between fields)
Grid gap:	gap-4 (16px) or gap-6 (24px)

6. Border Radius

Class	Value	Usage
rounded	4px	Badges, small elements
rounded-md	6px	Inputs, small buttons
rounded-lg	8px	Buttons, chips
rounded-xl	12px	Cards
rounded-2xl	16px	Large cards, modals
rounded-full	9999px	Circular avatars, pill badges

7. Shadows

Class	Usage
shadow-sm	Resting state of cards
shadow-md	Card hover state
shadow-lg	Modals, dropdowns
shadow-card	Custom card shadow (0 2px 8px rgba(15,23,42,0.08))
shadow-cardHover	Custom hover shadow

Standard card shadow pattern:

```
<div className="shadow-sm hover:shadow-md transition-shadow">
```

8. Buttons

Five variants:

Variant	Background	Text	Usage
primary	bg-primary-600	white	Main actions (View Details, Login)
secondary	bg-secondary-600	white	Secondary actions
danger	bg-danger-600	white	Destructive actions (Delete)
outline	transparent	text-primary-600	Alternative actions
ghost	transparent	text-slate-600	Subtle actions

Sizes:

- **sm** — compact buttons in tables
- **md** — standard form/action buttons
- **lg** — hero CTA buttons

9. Badges

Used extensively for medicine status, user roles, and categories.

Variant	Background	Text	Typical Use
success	bg-success-100	text-success-700	In Stock
danger	bg-danger-100	text-danger-700	Out of Stock
warning	bg-warning-100	text-warning-700	Limited Stock
info	bg-info-100	text-info-700	Jan Aushadhi
primary	bg-primary-100	text-primary-700	Nearby count
secondary	bg-secondary-100	text-secondary-700	Generic
neutral	bg-slate-100	text-slate-600	Category

10. Form Design

Consistent form styling across all pages:

Input states:

Default: border-slate-200
 Focused: border-primary-500 ring-2 ring-primary-200
 Error: border-danger-400 ring-2 ring-danger-200
 Disabled: bg-slate-50 text-slate-400 cursor-not-allowed

Error message: text-xs text-danger-600 below the input

Label: text-sm font-medium text-slate-700 above the input

11. Cards

Standard card construction:

```
<div className="
  bg-white                // White background
  rounded-xl              // 12px corners
  border-slate-100        // Subtle border
  shadow-sm               // Resting shadow
  hover:shadow-md         // Deeper shadow on hover
  hover:-translate-y-0.5  // Slight lift on hover
  transition-all duration-200 // Smooth transition
  p-4                     // Inner spacing
">
```

Best Value card (premium state):

```
<div className="border-primary-400 ring-2 ring-primary-200">
```

12. Icons

All icons come from two icon sets via [react-icons](#):

Set	Prefix	Style	Usage
Herolcons v2	HiOutline*	Outline	Navigation, UI actions
Herolcons v2	Hi* (solid)	Filled	Active/selected states
Material Design	Md*	Filled	Healthcare icons (MdMedication)

Accessibility rule: All decorative icons have `aria-hidden="true"`. Icons that convey meaning have `aria-label` on the parent element.

13. Animations

Animation	CSS	Usage
Page enter	page-enter class	All layout <main> elements
Card hover lift	hover:-translate-y-0.5	All cards
Smooth transitions	transition-all duration-200	Interactive elements
Skeleton pulse	animate-pulse	Loading states
Spinner	animate-spin	Loading indicators

14. Dark Mode Architecture

Dark mode is managed by:

1. `ThemeContext` — stores theme state in `localStorage`
2. `ThemeContext` applies/removes `dark` class on `<html>`
3. CSS variables in `index.css` override under `.dark {}`

4. Components use `dark:` Tailwind prefix for dark variants

```
{/* Component example */}  
<div className="bg-white dark:bg-slate-900 text-slate-900 dark:text-slate-100">
```

15. Z-Index Layers

Layer	Value	Used For
dropdown	100	Dropdowns, autocomplete
sticky	200	Sticky headers, CompareBar
overlay	300	Modal backdrop
modal	400	Modal content
toast	500	Toast notifications
tooltip	600	Tooltips

This layering is defined in `tokens.js` and documented for future reference.

16. Why Design Consistency Matters

In a healthcare application, visual inconsistency creates:

- **Confusion** — users don't know what to click
- **Distrust** — inconsistent design feels unprofessional
- **Cognitive load** — users have to re-learn each page

With a design system:

- Every button looks and behaves the same
- Users intuitively understand the interface
- Adding new pages automatically looks consistent
- The project looks professional for university evaluation